

17 January 2026

BrainEcho Lab
Erasmus Medical Centre
Dr. Molewaterplein 40,
Faculty Building,
Rotterdam,
The Netherlands

Dear Prof Kruizinga,

Re: "Postdoctoral Researcher in 4D Ultrasound for Image-Guided Neurosurgery"

I am applying for the post-doctoral position, having seen the opportunity on LinkedIn, as I believe I would be a ideal candidate for this role.

I am an applied mathematician by training, I have worked, as a computational scientist, at the interface between academic research and industry for over ten years, in the United Kingdom and Germany. My primary field is in therapy and imaging with medical ultrasound,

The position is of interest as I have been playing a significant role in the AEGEUS project, a EIC Pathfinder grant, dedicated to functional EEG and transcranial ultrasound imaging of the brain. This is based on changes in bulk speed of

Furthermore, I have worked in inter-disciplinary teams and can understand clinical problems from engineering, physics and mathematical perspectives. I have used the results of my simulations to deliver design recommendations to SMEs and international companies in the medical field in the context of foetal and doppler ultrasound, microwave ablation and in the design of ultrasound breast imaging devices. Although primarily a computational scientist I am familiar with data acquisition and processing from research scanners from Verasonics and us4us, as well as Fraunhofer devices.

I have used my mathematical skills and experience to write high-performance, well-documented, well tested, clinically deployable code, compliant with technical standards, such as ISO 13485 and IEC 62304. I have over ten years experience in python, Matlab and C++, as well as GPU knowledge both at a high level through cupy as well as low level programming of kernels in OpenCL.

I have proven expertise in acoustics (I am a member of IEC technical committee 87 on ultrasonics),

I am a maintainer of a ultrasound solver, k-wave-python, which is widely used around the world by both research and commercial groups.

I am applying as there are structural changes at my current role which may reduce opportunities for development. I am enthusiastic about this role as it clinical impact

Please find enclosed my résumé, and let me know if you require any additional information.

Yours faithfully,



David Sinden

Attached: Résumé